

1. Motivation & Introduction

The **Internet of Bio-Nano Things (IoBNT)** envisions nanomachine networks embedded inside biological environments for real-time monitoring and actuation. Molecular Communication (MC) is the canonical paradigm, yet current theoretical models rest on two problematic assumptions:

- **SISO / structured topology:** Most works study point-to-point or simple geometric MIMO links, unrepresentative of randomly distributed nanomachines in living tissue.
- **Super-capable individual agents:** Models assume perfect synchronisation, high-precision molecule counting, and complex on-board signal processing, far beyond current nanotechnology.

Biological insight: The cortex does not compute because neurons are individually intelligent. It computes because *millions of simple threshold-firing units act collectively.*

“Super-capable individual agents”



“Network of simple, numerous agents”

2. System Model

Setup: M spherical transceiver agents (radius a_{rx}) distributed via a Poisson point process inside a 3-D volume. Molecular motion follows *free diffusion* with homogeneous, isotropic coefficient D ; time is slotted into intervals T_s . *Fully absorbing* receivers are used for physical plausibility: surface absorption leaves the interior volume available for molecule storage, local computation, and energy harvesting.

Cumulative absorption probability that a molecule from agent j reaches agent i by time t :

$$g_{ij}(t) = \frac{a_{rx}}{d_{ij}} \operatorname{erfc}\left(\frac{d_{ij} - a_{rx}}{\sqrt{4Dt}}\right), \quad d_{ij} = |\mathbf{r}_i - \mathbf{r}_j|. \quad (1)$$

The per-interval absorption probability is $P_{ij}[\ell] = g_{ij}(\ell T_s) - g_{ij}((\ell-1)T_s)$, forming the symmetric **connectivity matrix** $\mathbf{P}[\ell]$ whose (i, j) -th entry encodes the coupling strength between agents i and j at lag ℓ . Diagonal entries are zero (no self-interaction).

Gaussian arrival model. Because each molecule is independently absorbed with probability $P_{ij}[\ell]$, the count is binomially distributed; for large N_0 this is well approximated as Gaussian. Conditioned on the binary state vector $\mathbf{x}[k] \in \{0, 1\}^M$ over the channel memory horizon L :

$$N_i^{(R)}[k] | \mathbf{x}[k-L : k-1] \sim \mathcal{N}(\mu_i[k], \sigma_i^2[k]), \quad (2)$$

$$\mu_i[k] = N_0 \sum_{\ell=1}^L \sum_j P_{ij}[\ell] x_j[k-\ell], \quad (3)$$

$$\sigma_i^2[k] = N_0 \sum_{\ell,j} P_{ij}[\ell](1 - P_{ij}[\ell]) x_j[k-\ell]. \quad (4)$$

The memory length L introduces inter-symbol interference (ISI), as molecules emitted in previous intervals are absorbed in later slots. This couples the current network state to its history; while typically an obstacle in MC, this feature could enable the system to integrate past states into its collective computations.

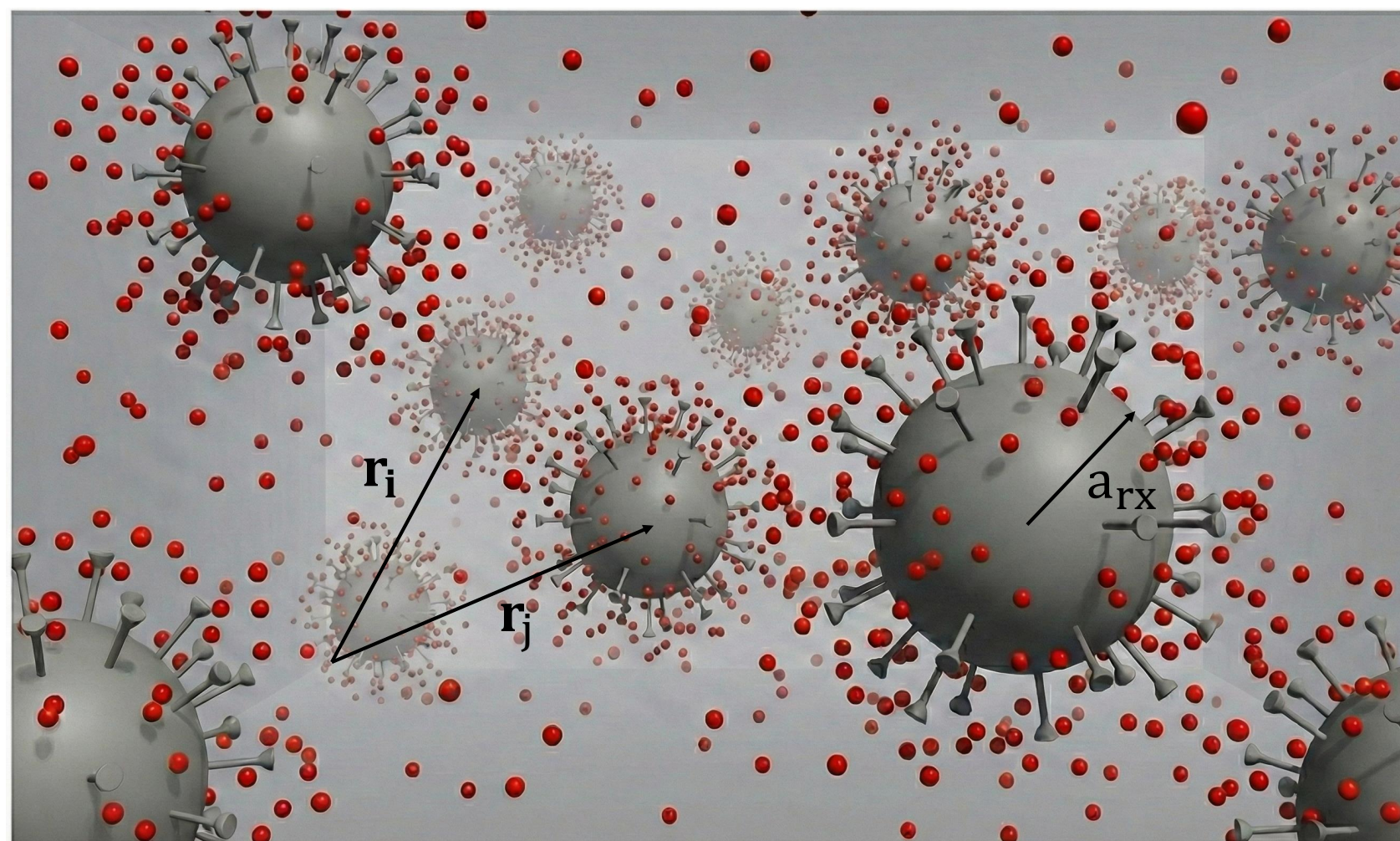
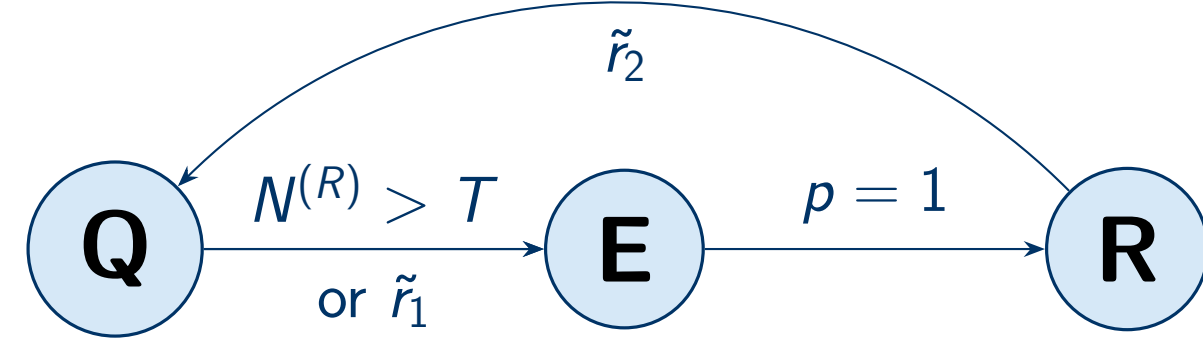


Fig. 1: Proposed swarm of excitable nanomachines communicating via diffusion.

3. Greenberg–Hastings Agent Model

We adopt the **Greenberg–Hastings (GH) cellular automaton** to model agent dynamics. Each agent occupies one of three discrete states:



Quiescent (Q) Resting; fires if $N^{(R)} > T$, or spontaneously w.p. $\tilde{r}_1$.

Excited (E) Releases N_0 molecules; transitions to R w.p. 1.

Refractory (R) Recovery; returns to Q w.p. $\tilde{r}_2$.

Excitation probability at step k for agent i :

$$\gamma_i[k] = \Pr\{N_i^{(R)}[k] > T\} = 1 - \Phi\left(\frac{T - \mu_i[k]}{\sigma_i[k]}\right), \quad (5)$$

where $\Phi(\cdot)$ is the standard normal CDF.

State-evolution equations derived from the GH transition rules:

$$r_i[k+1] = e_i[k] + (1 - \tilde{r}_2) r_i[k], \quad (6)$$

$$e_i[k+1] = (\tilde{r}_1 + \gamma_i[k]) q_i[k]. \quad (7)$$

The unconditional variance of the received molecule count separates into two physically distinct contributions:

$$\operatorname{Var}[N_i^{(R)}[k]] = \underbrace{\Sigma_i^{(1)}[k]}_{\text{random molecular arrivals}} + \underbrace{\Sigma_i^{(2)}[k]}_{\text{stochastic GH excitations}} \quad (8)$$

$$\Sigma_i^{(1)}[k] = \sum_{j,\ell} N_0 e_j[k-\ell] P_{ij}[\ell] (1 - P_{ij}[\ell]), \quad (9)$$

$$\Sigma_i^{(2)}[k] = \sum_{j,\ell} N_0^2 P_{ij}^2[\ell] e_j[k-\ell] (1 - e_j[k-\ell]). \quad (10)$$

The first term $\Sigma^{(1)}$ captures **diffusion noise** — binomial fluctuations in whether individual molecules reach their target. The second term $\Sigma^{(2)}$ captures **excitation noise** — the stochastic on/off firing of the Greenberg–Hastings agents across the swarm. Both terms are carried through the mean-field analysis to form the steady-state variance σ^{*2} .

4. Mean-Field Steady-State Analysis

As $k \rightarrow \infty$, state probabilities converge to $\{e_i^*, r_i^*, q_i^*\}$. Invoking mean-field independence ($e_j[k] \approx e_j^*$), the **unconditional mean and variance** become:

$$\boldsymbol{\mu}^* = N_0 \left(\sum_{\ell=1}^L \mathbf{P}[\ell] \right) \mathbf{e}^*, \quad (11)$$

$$\sigma^{*2} = N_0 \sum_{\ell=1}^L (\mathbf{P}[\ell] \odot (1 - \mathbf{P}[\ell])) \mathbf{e}^* + N_0^2 \sum_{\ell=1}^L \mathbf{P}^2[\ell] \mathbf{e}^* \odot (1 - \mathbf{e}^*), \quad (12)$$

where $\odot$ denotes element-wise multiplication.

These yield the M -dimensional fixed-point problem:

$$\mathbf{e}^* = \frac{\tilde{r}_1 + \Gamma(\mathbf{e}^*)}{1 + (\tilde{r}_1 + \Gamma(\mathbf{e}^*))(1 + 1/\tilde{r}_2)}, \quad (13)$$

$$\Gamma(\mathbf{e}^*) = 1 - \Phi\left(\frac{T - \boldsymbol{\mu}^*}{\boldsymbol{\sigma}^*}\right). \quad (14)$$

Solved via **under-relaxed fixed-point iteration**: $\mathbf{e}_{k+1}^* = \lambda g(\mathbf{e}_k^*) + (1 - \lambda)\mathbf{e}_k^*$, $\lambda = 0.9$, $\varepsilon = 10^{-12}$.

Validation against stochastic simulation (100 runs, 500 steps each per threshold value):

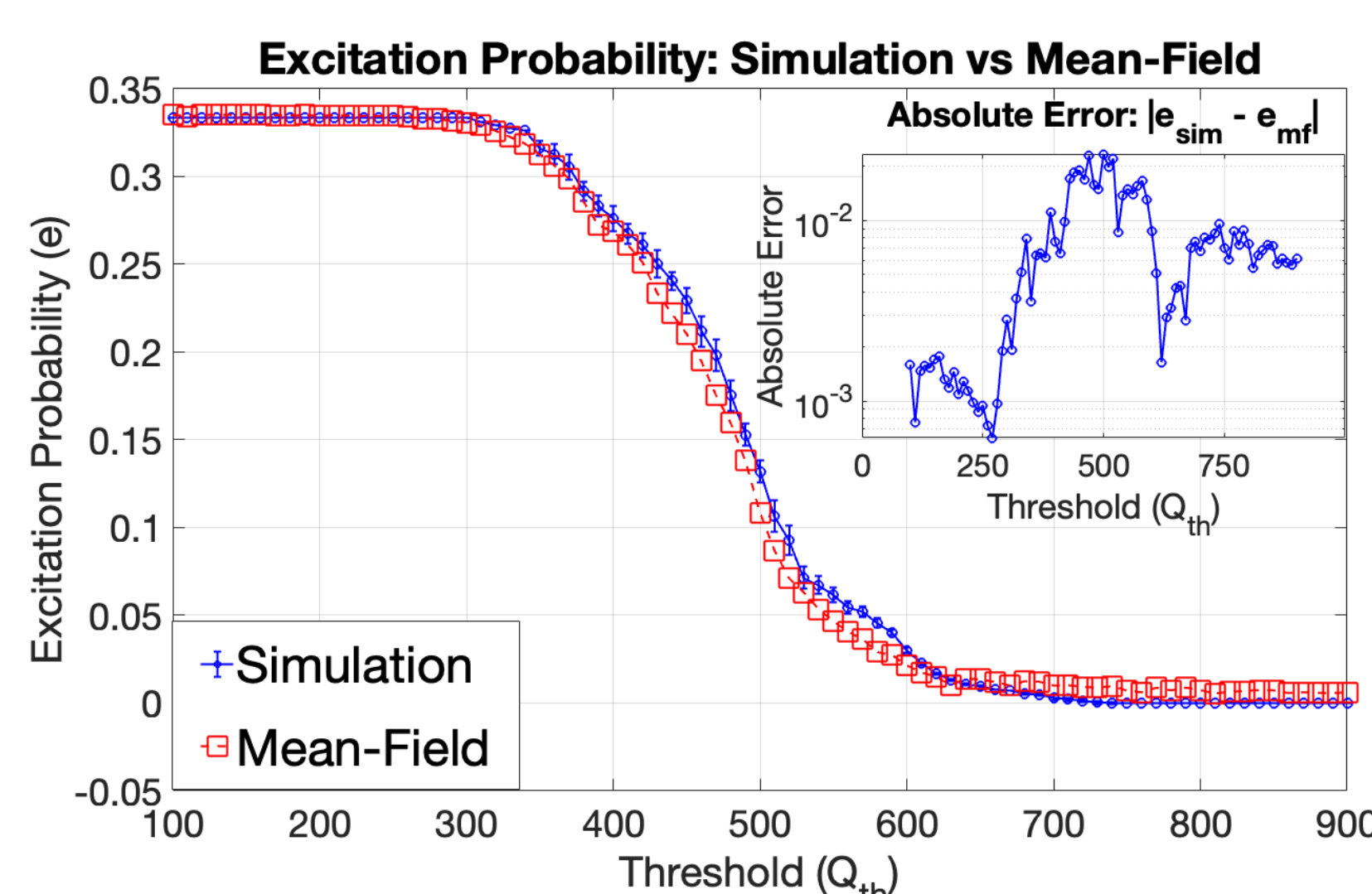


Fig. 4: Theoretical (mean-field) vs. simulated excitation probability. Inset: absolute error (log scale). Divergence at T_c is physically expected — long-range inter-agent correlations break the independence assumption exactly at the critical point.

5. Results: Phase Transition & Critical Regime

Simulation parameters:

Parameter	Value	Parameter	Value
Agents (N)	100	Radius (a_{rx})	4 μm
Medium radius	20 μm	Diff. (D)	79.4 $\mu\text{m}^2/\text{s}$
Emission (N_0)	100 molec.	Memory (L)	5
Time step (T_s)	1 s	$\tilde{r}_2$	0.5

As Q_{th} increases, mean activity drops sharply near $Q_{th} \approx 550$. The **standard deviation peaks at the transition**, indicating maximum network susceptibility — the hallmark of a critical regime:

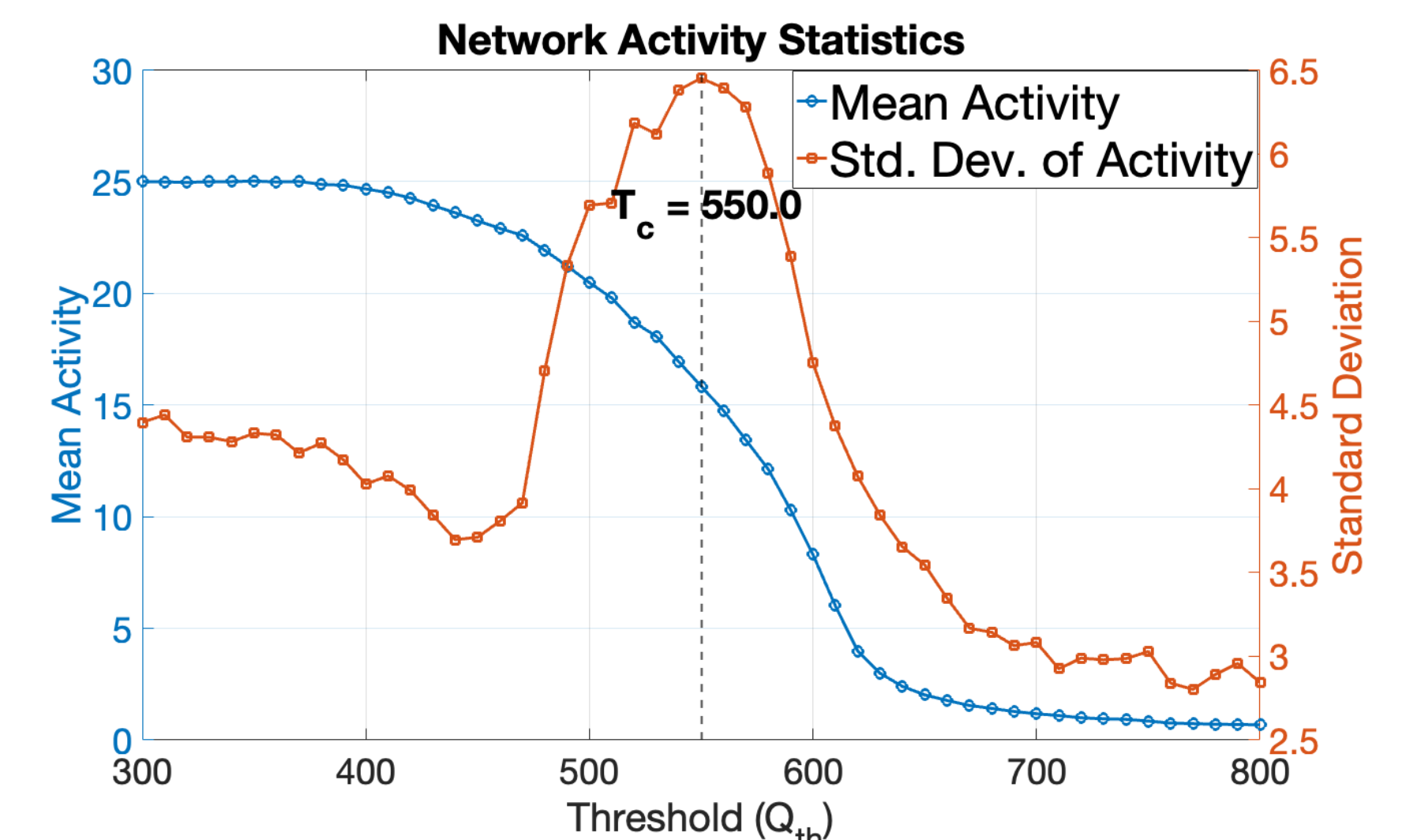


Fig. 2: Mean and std. dev. of excited activity vs. activation threshold Q_{th} .

Regime	Threshold	Behaviour
Sub-critical (saturation)	$Q_{th} < 450$	Noise dominates
Critical (optimal)	$Q_{th} \approx 500\text{--}550$	Max. info. transfer
Super-critical (silence)	$Q_{th} > 600$	Activity dies out

6. Results: Mutual Information at Criticality

MI metrics quantify statistical coupling across the swarm:

$$\bar{I}_{\text{pair}} = \binom{N}{2}^{-1} \sum_{i < j} I(X_i; X_j), \quad \bar{I}_{\text{coll}} = \frac{1}{N} \sum_{i=1}^N I(X_i; M_i), \quad (15)$$

where M_i is the aggregate activity of all agents excluding i . Both metrics **peak precisely at** $T_c = 550$:

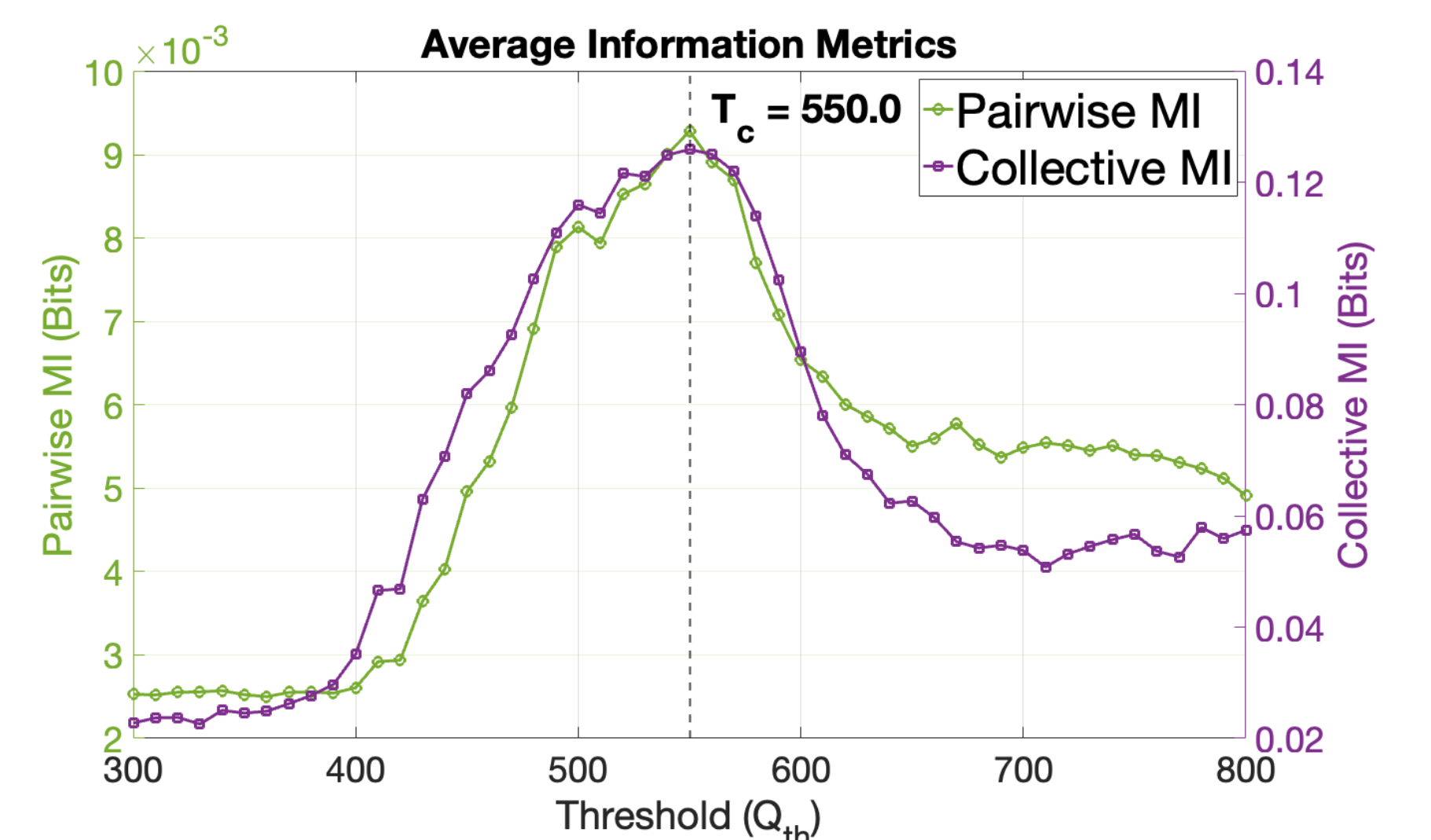


Fig. 3: Pairwise and collective MI peak at $T_c=550$, confirming that the critical regime maximises collective information processing.

The co-location of the MI peak with the susceptibility peak (Fig. 2) and the mean-field breakdown (Fig. 4) constitutes a three-way signature of **criticality** — consistent with findings from cortical network studies in neuroscience.

7. Conclusions & Future Work

- A **decentralised swarm** of simple nanomachines can achieve rich collective dynamics via diffusion alone.
- Tuning the **activation threshold** drives the network to a critical regime that maximises information propagation.
- A **closed-form mean-field fixed-point framework** accurately predicts macroscopic behaviour outside the transition.
- **MI metrics peak at criticality**, providing a quantitative signature of collective intelligence.

Future Directions:

- Systematic **parameter sensitivity analysis** (density, D , geometry).
- External input-output channel model for rigorous **capacity analysis**.
- Analytical framework to **account for inter-node correlations** near T_c (finite-size scaling).
- Relax isotropic diffusion for **heterogeneous in-body tissue**.